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Chapter 1

Introduction

1.1 Problem Domain

In multi-agent systems (MAS), agents often need to collaborate or compete with each other to achieve specific goals. A critical aspect of this collaboration is communication, as agents must share information and coordinate actions efficiently. The emergence of communication in MAS, especially in zero-shot scenarios where agents must interact with novel partners, remains a challenging and unexplored area. Most current approaches either rely on predefined communication protocols or expect agents to develop language during training. However, the generalization of these emergent languages to new agents, without retraining or adapting, has shown limited success.

Our research focuses into the problem of developing emergent languages that are transferable across agents, without introducing bias into their policies. The study aims to fill the gap between the emergence of communication and its scalability across independent agents.

1.2 Research Problem Statement

Despite progress in emergent communication, existing methods often face challenges when generalizing the learned communication protocols to new agents. In many cases, the emergent languages are either too specific to the agents' training environment or require modifications to the agents' policies, limiting scalability and interoperability. Furthermore, the emergence of multiple languages among different agent pairs can prevent effective zero-shot coordination, as agents may not be able to understand or predict the actions of unfamiliar partners.

This research aims to address the following problem: *How can a common emergent lan-*

guage be developed among independent agents without changing to the agent's underlying policies?

The study will explore how agents can develop a transferable and generalizable communication system. The proposed approach focuses on language emergence and transferability, ensuring that new agents can adopt the common language without altering their policy structure, promoting both scalability and interoperability in multi agent systems.

Chapter 2

Literature Review

2.1 Related Research

Following researches focused on either generalizing policy to communicate or collaborate in zero shot settings or better focus on emergence of communication in multi agent systems.

2.1.1 Quasi-Equivalence Discovery [1]

In Quasi-Equivalence Discovery (QED) for zero-shot communication (ZSC), quasi-equivalence refers to the identification of symmetries in communication protocols. Symmetries refer to situations where different actions, messages, or sequences of actions lead to the same outcome or result. These symmetries allow agents to generalize their learned communication strategies to independently trained agents, even in zero-shot scenarios where the agents have not been trained together.

2.1.1.1 Summary of the research item

Quasi-Equivalence Discovery (QED), iteratively discovers equivalence mappings between optimal policies in multi-agent systems. Unlike traditional self-play, where agents co-adapt during training, QED focuses on finding policies that generalize on different agents.

The algorithm first identifies symmetries by training agents with different policies and observing how different action or message sequences can achieve the same result. It does this by comparing the behavior of multiple agents and finding equivalent policies that lead to the same goal. After identifying these equivalences, QED trains new sets of policies based on these symmetry mappings. This allows the agents to communicate successfully with novel agents, because they can recognize the symmetries and generalize their

communication.

2.1.1.2 Critical analysis of the research item

The task and environment selected in paper is creating a message and predicting the goal based of that message. They are using equivalence mapping and training new joint optimal policies and then using new optimal policy, updating set of equivalence mappings. After iteratively training, they select one unique policy which is capable of zero shot communication. Training continues until cross play score is almost equivalent to self play.

2.1.1.3 Relationship to the proposed research work

While QED focuses training multiple different policies and then finding a joint policy that generalizes for zero shot communication with independent agents, our work focuses on emerging a language in agents and transfer of that language to new incoming agents without changing or creating bias in their policy.

2.1.2 Multi-lingual agents through multi-headed neural networks [4]

In the paper, the authors maintained multiple heads in the neural network architecture for each agent, where each head corresponds to a distinct communication language that the agent learns with different partners.

2.1.2.1 Summary of the research item

Both the speaker and listener agents are modeled using deep neural networks. The core idea is to assign a separate output head for each possible conversational partner. The shared layers of the neural network are used for feature extraction. It is assumed that each agent when communicating with other agent have full knowledge about their identity.

The heads are isolated from each other, meaning when training occurs with one partner, only the corresponding head is updated, preventing other heads from being affected from catastrophic forgetting.

2.1.2.2 Critical analysis of the research item

A key limitation noted by the authors is that the number of languages grows quadratically with the number of agents. If many agents are involved, maintaining separate heads for each one may lead to memory and computational overhead, not allowing scalability. Furthermore, the method assumes that agents know who their conversational partner is, which

allows them to select the correct output head. In more realistic or dynamic environments, agents may not have this information reducing the scalability even more.

2.1.2.3 Relationship to the proposed research work

The objectives of both researches are same i.e mitigating the problem of different languages emerging for every independent agent and thus preventing zero short coordination. However, our work focuses more on making language common for every agent rather than training agents to learn multiple language thus allowing scalability and further research on topic.

2.1.3 Towards Human-Agent Communication via the Information Bottleneck Principle [6]

In traditional emergent communication, which focuses on optimizing task-specific utility, can lead to communication that is either too complex for humans to understand or inefficient in terms of convergence rates. To address this, they explore how a trade off between utility, informativeness, and complexity can help agents develop more human-like communication.

2.1.3.1 Summary of the research item

The goal of paper is to create an emergent communication system where agents not only maximize task performance but also optimize the informativeness and complexity of their communication signals.

They are using information bottleneck framework to balance the trade off between information and complexity of messages. For this they are using VQ-VIB, that combines elements from vector quantized variational autoencoders (VQ-VAE) and B-variational autoencoders, to compress and discretize communication signals. The overall loss function includes terms for utility, informativeness, and complexity. The agents are trained to maximize this objective function.

2.1.3.2 Critical analysis of the research item

Paper explores another dimension in emergence of communication between multi agent systems where the emergence is inspired by human like communication for effective and better convergence. Experiments of these papers indeed show better convergence than vanilla methods for emergence of communication.

2.1.3.3 Relationship to the proposed research work

While our work focuses on emergence and transfer of language in multi agent systems, this paper focuses on the former module i.e efficient emergence of language in multi agent systems using VQ-VIB.

2.1.4 Emergent Discrete Communication in Semantic Spaces [5]

This paper explores how agents can develop discrete communication tokens embedded in a continuous semantic space, which improves both task performance and flexibility while also supporting human understanding.

2.1.4.1 Summary of the Research Item

The paper trains agents to develop discrete communication tokens within a continuous semantic space to communicate effectively in multi-agent tasks. The sender uses a fixed set of tokens learned during training, but these tokens capture semantic relationships, allowing the agents to generalize and achieve zero-shot communication.

2.1.4.2 Critical Analysis of the Research Item

without ground truth labels or shared pre trained embedding, agents lack a common semantic framework to understand each other's tokens. They might create their own unique latent space based on their training, which can be vastly different from what other agents might interpret. If two agents develop distinct interpretations of similar concepts they cannot effectively communicate about that concept.

2.1.4.3 Relationship to the Proposed Research Work

While our proposed work focuses on the emergence and transfer of communication in multi-agent systems, this paper contributes to the efficient emergence of discrete communication by using prototype-based learning with semantically structured tokens. Our goals aligns with the paper's goals of developing generalizable communication protocols, but differs by placing greater emphasis on language transfer between agents.

2.1.5 Collaborating with Humans without Human Data [3]

paper proposes Fictitious Co-Play(FCP) where agents are capable of zero shot coordination with humans without need of humans behavioural data for training.

2.1.5.1 Summary of the research item

this papers address the problem of training Agents to collaborate with humans without any training on behavioural data by human with zero shot coordination. tradition methods either use data of human behaviour or self play techniques but they do not generalize to new humans. this technique of Fictitious Co-Play(FCP) allow agent to play different self play agent with different behaviours and skill level to develop a generalized policy to coordinate with new partners(humans) even they interacted for the first time.

2.1.5.2 Critical analysis of the research item

this paper is a cost efficient way for zero shot coordination with new agents or we can say humans without being trained on humans behavioural data. Its generalization increases with how many agents we interacted during training thus reducing scalability to large problem.

2.1.5.3 Relationship to the proposed research work

This paper focuses on zero shot coordination but it is behavioural dependent and ignores communication aspect. Our goal aligns zero-shot coordination but our assumption adds zero shot coordination can be achieved through common language also allowing better scalability and interoperability.

2.1.6 Compositionality and Capacity in Emergent Languages [2]

In this Paper, author tried to investigate how capacity of neural network and bandwidth of communication affect the emergence of compositional languages in multi-agent systems.

2.1.6.1 Summary of the research item

Authors found that low capacity networks can avoid over fitting and hence develop more composed structure of language and same is the case with bandwidth of message as more composed the message is less bandwidth it requires. Their results suggest that right balance between bandwidth and capacity helps us get more human-like composed structures in language.

2.1.6.2 Critical analysis of the research item

This paper is good for much composed language this composed language can help us build a better language but this this language can have restriction of evolution as message size is just to facilitate our correct information. but we want more information to store in this amount of message vector which is not possible.

2.1.6.3 Relationship to the proposed research work

This paper is good for much composed language this composed language can help us build a better language and transfer it to other agents which will be much more organized.

2.2 Analysis Summary of Research Items

Research Item	Critical Analysis	Relation to Proposed Work
Quasi-Equivalence Discovery (QED)	Focuses on policy symmetry mappings for zero-shot communication. Limited exploration of language emergence.	Useful for policy generalization; our work extends to emergent language transfer.
Multi-lingual agents through multi-headed neural networks	Scalability issues due to the quadratic growth of language heads, assumes agents know their communication partner.	Our work aims to create a common language, avoiding multiple heads and ensuring scalability.
Towards Human-Agent Communication via Information Bottleneck	Focuses on optimizing communication for human understanding but is constrained by task-specific utility.	Offers insights into optimizing informativeness and complexity of language, aligning with our work on emergent language.
Emergent Communication in Multi-Agent Systems: A Prototype-Based Approach	Lacks a shared semantic framework, potentially creating divergent interpretations among agents.	We focus on the transfer of a shared language between agents, addressing the problem of divergent communication tokens.
Collaborating with Humans without Human Data	Uses fictitious co-play for zero-shot coordination with humans, but neglects the role of communication.	Our work includes communication as a key aspect of zero-shot coordination, ensuring better scalability.
Compositionality and Capacity in Emergent Languages	Trade-off between bandwidth and language compositionality may restrict information capacity in communication.	We aim to find a balance between compositionality and transferability of the language in a scalable way.

Table 2.1: Analysis Summary of Research Items

Chapter 3

Proposed Approach

There has been significant progress in the field of agents learning to develop languages for communication to accomplish tasks. However, the challenge of zero-shot communication remains largely unaddressed. While some studies have attempted to solve this problem, the majority focus on policy learning rather than actual language transfer. One notable paper utilized quasi-equivalence to tackle this issue, but a direct method for teaching agents a shared language for zero-shot communication is still lacking, as quasi-equivalence needs to be clearly defined before enabling effective agent interaction.

Objective

Our proposed approach focuses on developing a common language that allows agents to first create their own language and then transfer this language to other agents. This will be achieved by analyzing how agents communicate their ideas and send messages to convey their plans effectively.

Methodology

To implement this, we will introduce a penalty term in the loss function, leveraging a trained agent to guide the learning process. This penalty will encourage agents to adhere to a communication protocol, ensuring they convey their ideas in a consistent language rather than random messages.

Policy Independence

The goal is to maintain policy independence while establishing a shared communication protocol. Our algorithm will enable a teacher agent to critique a child agent based on the policy the new agent is following, rather than solely relying on the original policy it was trained on.

Significance

This approach aims to facilitate meaningful communication among agents in zero-shot scenarios, enhancing their collaborative capabilities while preserving their individual learning processes. By developing a robust framework for language transfer, we hope to bridge the gap between policy learning and effective communication, thereby advancing multi-agent systems.

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